

The Stochastic Selector: Exact Rational Belief-State Safety Values under Partial Observation

Edition 2, revised after a joint audit

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Abstract

The viable-selector framework reads every obstruction certificate as the emptiness of a response correspondence at an information state; its closing section previewed the stochastic counterpart, in which the information state is a distribution and viability is the value-one level of a belief-state safety recursion. This paper develops that theory. The belief-state safety value is the stochastic selector, and its finite-horizon values are piecewise linear with exactly representable witnesses: over rational input data, the backward recursion propagates rational alpha-vectors, so all values are exact rationals computable without floating point. The model is stated on the augmented state space with total transition rows, an absorbing unsafe state carrying its own observation, and masked witnesses. At the deterministic limit the recursion is replaced by a distinct robust operator — a survived-mass formula over blind action sequences — which recovers the robust verdicts of the companion papers and implies their deficit bounds. Obstruction certificates are read systematically as deficit lower bounds: the chance-constrained common-action bound, the degenerate minimal-mass bound, and monotonicity. On the audited delayed-regime class the theory is carried exactly: the class-restricted alpha-vector recursion reproduces the closed-form values on all forty-eight audited cells, the certificate partition coincides with the value boundaries, the attaining witness and the value's jump loci trace the certificate boundary exactly, the deficit is constant on the certified region, and the class declaration is quantified — the declared hold class and the unrestricted sequential blind class coincide at the one-step horizon and differ exactly on the timing cells, where open-loop time-variation within the blind window lifts the value to one. All computations run in exact rational arithmetic.

1. Introduction

The companion papers develop obstruction certificates for viability under incomplete observation — finite witnesses that no observation-based policy is safe — and unify them as emptiness of a response correspondence at an information state (Abaee, 2026a,b,c). The unifying paper closes with the stochastic counterpart, left there as an outlook: replace the set-valued information state by a distribution,

the correspondence's nonemptiness by the value-one level of a safety recursion, and the certificate by a lower bound on the deficit from unity. This paper delivers that theory, seeded by the worked instance already carried in the calculus paper's probabilistic section and its coverage audit. This second edition incorporates the dispositions of a joint audit of the first: a policy-class parameter made explicit throughout the value theory, the model's boundary cases completed, the selector's attaining set separated from its value-one level, and the audit's own computations absorbed as verified results (Section 6 lists the dispositions' effects; the joint audit record accompanies the paper).

Four results. First, the *exact representation*: class-restricted belief-state safety values are piecewise linear in the belief at every finite horizon, and over rational data the witnessing alpha-vectors are rational and propagated exactly by the backward recursion (Section 3, Propositions 1 and 2) — no floating point enters the value theory, in keeping with the companions' verification discipline. Second, the *degeneration*: at the deterministic limit the stochastic operator is replaced by its robust counterpart, which collapses to a survived-mass formula over blind action sequences and recovers the companion papers' robust verdicts and their deficit bound (Proposition 2). Third, the *certificate reading*: obstruction certificates are deficit lower bounds, systematically (Proposition 3). Fourth, the *agreement theorem* on the audited classes: on the delayed hidden-regime grid the class-restricted alpha-recursion, the closed form, the certificate partition, and the witness-family boundaries coincide cell by cell, with a constant deficit on the certified region and a quantified class declaration (Section 5, Theorem 2). Everything is verified in exact rational arithmetic (Section 6).

2. The stochastic selector

The augmented state space. Let the physical safe set be $\mathcal{V}$ and let the state space of the model be $X^\perp = \mathcal{V} \cup \{\perp\}$, where $\perp$ is an absorbing unsafe state. For $x \in \mathcal{V}$ the transition row splits between $\mathcal{V}$ and $\perp$,

$$\sum_{x' \in \mathcal{V}} T(x' | x, a) \leq 1, T(\perp | x, a) = 1 - \sum_{x' \in \mathcal{V}} T(x' | x, a),$$

and the bottom row is $T(\perp | \perp, a) = 1$: every row of T sums to one and $\perp$ is absorbing, so exit from $\mathcal{V}$ is absorption and no

trajectory re-enters $\mathcal{V}$ from outside it. The observation model is completed on all of $X^\perp$: there is a distinguished observation $y_\perp$ with $g(y_\perp \mid \perp, a, \perp) = 1$ (and $g(y \mid \cdot, \cdot, \perp) = 0$ for $y \neq y_\perp$), while $g(\cdot \mid x, a, x')$ on $\mathcal{V}$ is arbitrary non-negative with the usual normalization per (x, a) . A belief is a distribution b over $X^\perp$, updated by Bayes' rule $\tau(b, a, y)$, with $\mathbb{P}(y \mid b, a) = \sum_{x, x'} b(x) T(x' \mid x, a) g(y \mid x, a, x')$; because every row sums to one and every (x, a, x') emits exactly one observation with probability one, the laws $\mathbb{P}(y \mid b, a)$ sum to one over y at every belief — no sub-probability bookkeeping is needed. The *safety value* at horizon k , for a declared policy class Π , is the maximal class-survival probability

$$V_k^\Pi(b) = \max_{\pi \in \Pi} \mathbb{P}_\pi(x_t \in \mathcal{V} \text{ for } t = 0, \dots, k \mid b),$$

the horizon counting the current state. The unrestricted class $\Pi = \Pi_{\text{seq}}$ is the class of all sequential policies.

The selector and its levels. In the selector framework this is the distributional counterpart of the recursive correspondence: the information state is b , a response is a class-admissible policy segment. Two objects must be distinguished. The *attaining set* at b ,

$$\Gamma_k^s(b) = \left\{ a \in A : \sum_y \mathbb{P}(y \mid b, a) V_{k-1}^\Pi(\tau(b, a, y)) = V_k^\Pi(b) \right\},$$

collects the actions attaining the value; for finite A it is nonempty at every belief and horizon — nonemptiness is automatic and carries no viability content. The *value-one level*,

$$S_k^\Pi(b) = \left\{ a \in A : \sum_y \mathbb{P}(y \mid b, a) V_{k-1}^\Pi(\tau(b, a, y)) = 1 \right\},$$

is the selector's viable-response set, and its nonemptiness corresponds exactly to $V_k^\Pi(b) = 1$. Obstruction certificates correspond to deficits $1 - V_k^\Pi(b) > 0$. The first edition conflated the two nonemptiness statements; the dispositions repair the definition, and the distinction is verified instance by instance (Section 6).

Theorem 1 (belief-state value recursion). $V_0^\Pi(b) = b(\mathcal{V})$ (the horizon counts the current state), and for $k \geq 0$,

$$V_{k+1}^\Pi(b) = \max_{a \in A(\Pi)} \sum_{y \in Y} \mathbb{P}(y \mid b, a) V_k^\Pi(\tau(b, a, y)),$$

where $A(\Pi)$ is the class-admissible action set at the current stage for Markov classes Π . The recursion is exact on every finite horizon; over the unrestricted class it is the Smallwood–Sondik belief-state value iteration specialized to the absorbing safety reward (Smallwood and Sondik, 1973).

The proof is standard value iteration for the belief-state Markov decision process, with the absorbing state making V_k^Π the k -step class-survival probability. For non-Markov classes (the hold and open-loop classes of Section 5) the class constraint couples stages; the audited computations there use the equivalent segment form of Proposition 1 below.

3. Exact piecewise-linear values

Proposition 1 (rational alpha-vectors, class-restricted). Fix a declared class Π . At every horizon k , V_k^Π is piecewise linear and convex: $V_k^\Pi(b) = \max_{\alpha \in \Gamma_k^\Pi} \alpha^\top b$ for a finite witness set Γ_k^Π on $\mathbb{Q}^{|\mathcal{V}|+1}$ (coordinates indexed by $\mathcal{V} \cup \{\perp\}$), constructed by the masked backup

$$\alpha^{a, \gamma_\cdot}(x) = \sum_{x'} T(x' \mid x, a) \sum_y g(y \mid x, a, x') \gamma_y(x') \gamma_y \in \Gamma_k^\Pi,$$

with $\Gamma_0^\Pi = \{\mathbf{1}_\mathcal{V}\}$ and the boundary condition $\alpha(\perp) = 0$ applied at every backup (the backup acts on $\mathcal{V}$ only; the $\perp$ -coordinate of every witness is anchored at zero and remains zero under

the backup) and $\Gamma_{k+1}^\Pi = \{\alpha^{a, \gamma_\cdot} : a \in A(\Pi), \gamma_\cdot \text{ a selection}\}$, duplicates removed by exact equality. For a blind open-loop window the equivalent segment form applies: each admissible action segment $(u_0, \dots, u_m) \in \Pi$ contributes the witness $\alpha^{\text{seg}} = (\mathbf{1}[\text{the segment saves branch } x])_x$ and $V_k^\Pi(b) = \max_{\text{seg}} \alpha^{\text{seg}^\top} b$. Over rational T , g , and priors, every α is an exact rational vector, every $V_k^\Pi(b)$ at a rational belief is an exact rational, and the attaining witness is part of the computation. The worst-case growth is $|\Gamma_{k+1}^\Pi| \leq |A| |\Gamma_k^\Pi|^{|\mathcal{Y}|}$; deduplication and pruned selections are the practical controls, and the exponential law is the honest complexity statement for the general class.

Proof sketch. The PL structure is Smallwood–Sondik's; the masking factor anchors $\alpha(\perp) = 0$ (survival requires the current state in $\mathcal{V}$, and $\perp$ is absorbing, so no backup can revive a lost branch). The point here is the exact-arithmetic discipline: the backup maps rational vectors to rational vectors by finite sums and products, selections are finite, and equality tests for deduplication are exact on rationals. The segment form is the same statement with the class constraint folded into the enumeration of segments. $\square$

Remark 1 (the two operators). The expectation over the disturbance law and the adversarial reading of its support are *different Bellman operators*. Proposition 2 states the deterministic degeneration as a change of operator: when the kernels are deterministic the stochastic expectation is replaced by the robust (adversarial-support) reading, and the resulting value is the robust value of the companion papers. The degeneration is a theorem about the robust operator; it is not a claim about the stochastic plant, whose expectation and robust readings differ in general (Section 7).

Proposition 2 (deterministic-limit degeneration). Let the kernels be deterministic, $x^+ = F(x, a, d)$, and let the disturbance be read adversarially within its support — the robust operator of Remark 1. Let the window be blind with declared sequential class Π_B . Then the degenerated value is the *survived-mass formula*

$$V_k(b) = \max_{(u_0, \dots) \in \Pi_B} \sum_{x \in \text{supp } b} b(x) \mathbf{1}[x \text{ survives } (u_0, \dots)],$$

a maximum of rational linear functions of b (the surviving indicators are the degenerate alpha-vectors). Consequently: if the support B carries a jointly surviving blind sequence, $V_k(b) = 1$; and if every declared sequence loses some branch, then $V_k(b) \leq 1 - \min_{x \in B} b(x)$ — the robust common-action verdicts and the minimal-mass deficit bound of the companion papers, recovered as exact values of the robust operator.

Proof sketch. With deterministic kernels the posterior support follows the set-valued recursion, survival from a branch under a fixed sequence is a deterministic event, and prior mass sums over surviving branches; maximizing over declared sequences gives the formula, from which both consequences are immediate (a lost branch carries at least the minimal mass). $\square$

4. Certificates as deficit bounds

Write $\Delta_k(b) = 1 - V_k(b)$ for the deficit. For the statement below, W_k denotes the *k-step robust kernel in belief space*: the family of beliefs whose support admits a jointly surviving declared-class sequence (a subset of the belief simplex, not of the state space).

Proposition 3 (deficit lower bounds). Each companion certificate is a computable lower bound on the deficit: (i) *chance-constrained common action*: if the one-step violation probabilities satisfy $\sum_x b(x) p(x, a) \geq \delta$ for every $a \in A$, then $\Delta_k(b) \geq \delta$

for every $k \geq 1$; (ii) *degenerate minimal mass*: at the deterministic limit, if $b \notin W_k$ (the support admits no jointly surviving declared sequence), then $\Delta_k(b) \geq \min_{x \in \text{supp } b} b(x)$; (iii) *monotonicity*: $V_{k+1}(b) \leq V_k(b)$, so the deficit is nondecreasing in the horizon. For finite horizons the exact deficit $1 - \max_{\alpha \in \Gamma_k} \alpha^\top b$ is computable, and the bounds are witnesses of it, not surrogates.

Proof sketch. (i) is the chance-constrained obstruction of the companion paper: every action loses probability at least δ at the first step and survival probabilities never increase. (ii) is Proposition 2’s second consequence. (iii) holds since $V_{k+1} = \max_a \mathbb{E}[V_k \circ \tau] \leq \max_a \mathbb{E}[V_{k-1} \circ \tau] = V_k$ by induction. $\square$

5. Agreement on the audited classes

The delayed hidden-regime instance of the calculus paper — regimes $\theta \in \{-1, +1\}$, blind window T_{obs} , declared hold class $\{u \equiv \pm 1\}$, prior $(\frac{1}{2}, \frac{1}{2})$ from z_0 — is the agreement test bed.

Theorem 2 (certificate-value agreement). On the audited grid ($z_0 \in \{1.0, \dots, 2.5\}$, $T_{\text{obs}} \in \{1, 2, 3\}$; 48 cells), over the declared hold class: (a) the class-restricted alpha-vector recursion of Proposition 1 returns exactly the closed form

$$V_k(b_0) = \frac{1}{2}(\mathbf{1}[z_0 - k \geq 1] + \mathbf{1}[z_0 + k \geq 1]) \quad (k \leq T_{\text{obs}}),$$

$$V_k(b_0) = \frac{1}{2}(1 + \mathbf{1}[z_0 \geq 1 + T_{\text{obs}}]) \quad (k > T_{\text{obs}});$$

on the audited grid the second indicator of the first line is vacuous ($z_0 + k \geq 2$); it is retained for branch symmetry. (b) the certificate partition coincides with the value boundaries: “viable” iff $V_k \equiv 1$; common-action-certified iff $V_1(b_0) < 1$; timing-certified iff $V_1(b_0) = 1$ but $V_{T_{\text{obs}}+1}(b_0) < 1$; (c) at the one-step horizon the attaining witness family changes exactly at $z_0 = 2$: below it two distinct witnesses attain $\frac{1}{2}$; at and above it, the merged unit witness attains 1 — and at horizon $k \leq T_{\text{obs}}$ the family changes exactly at $z_0 = 1 + k$. Read as a function of the continuous parameter z_0 , the hold-class value has jump discontinuities exactly at $z_0 = 1 + \min(k, T_{\text{obs}})$ (a jump of $\frac{1}{2}$, from $\frac{1}{2}$ to 1); on the audited grid these loci are the certificate edges. (Kinks in the strict sense belong to the piecewise-linear value as a function of the belief, not of z_0 .) (d) the deficit equals $\frac{1}{2}$ on every nonviable cell — the minimal-mass bound attained with the minimal prior; and (e) the class declaration is quantified: the declared hold class and the unrestricted sequential blind class coincide at the one-step horizon; for $k \geq 2$ they differ exactly on the timing cells $\{2 \leq z_0 < 1 + T_{\text{obs}}\}$, where the unrestricted class attains 1 — a single action already saves both branches at $z_0 \geq 2$ (the down-branch sits at $z_0 - 1 \geq 1$), and the alternating open-loop sequence $u, -u, u, \dots$ keeps both branches within $[z_0 - 1, z_0 + 1]$ at every horizon. The timing cells are exactly the cells where open-loop time-variation within the blind window lifts the declared-class value.

Proof sketch. The two-regime blind POMDP has, at each window horizon, exactly two alpha-vectors $\alpha^u = (\mathbf{1}[z_0 - ut \geq 1 \ \forall t \leq \cdot], \mathbf{1}[z_0 + ut \geq 1 \ \forall t \leq \cdot])$, $u = \pm 1$; evaluating $\max_u \alpha^u b_0$ at the uniform prior gives (a), from which (b) and (d) read off against the companion paper’s certificate conditions; (c) compares the attaining witness family across the boundaries $z_0 = 2$ and $z_0 = 1 + k$, and the jump loci follow from the closed form’s indicators. For (e), the one-step coincidence is immediate (both classes choose one action), the alternating sequence’s branch positions are computed exactly (they oscillate between $z_0 \pm 1$ and z_0 , minimum $z_0 - 1$), and brute force over all sequences gives the matching upper bound. The complete computations are machine-verified in exact rational arithmetic. $\square$

Remark 2 (the timing obstruction is class-scoped). The calculus paper’s timing certificates ($\sigma^*(B_0) = z_0 - 1$, $T_{\text{obs}} < \sigma^*$) are statements about the declared hold-until-review class — the institutional reading of a sampled-review regulator, which is how that paper’s blind-window control classes are defined. Theorem 2(e) quantifies the contrast: within the declared class the timing cells are nonviable with deficit exactly $\frac{1}{2}$; under the unrestricted sequential blind class the same cells attain value 1 by oscillation, and no other cell differs. The timing obstruction therefore certifies exactly what it claims — nonviability *within the declared class* — and no statement of the companion papers is altered by this reading. The word “adaptive” is avoided for blind windows: while the window is blind, no observation updates the belief, and the only distinction available is hold-constant versus time-varying open-loop control.

6. Verified re-derivations

`paper2_stochastic_selector_v2_verify.py` (standard library; exact rational arithmetic; deterministic) re-derives every claim in fifteen check families, all passing:

- **S1** two-floor instance: the backup produces exactly $\Gamma_1 = \{(1, 0), (0, 1)\}$ on the physical coordinates, $V_k(b_0) = \frac{1}{2}$ for all probed horizons, the chance bound $\delta = \frac{1}{2}$ attained;
- **S2** piecewise linearity confirmed at eleven simplex beliefs against the direct value $\max(b_1, b_2)$;
- **S3** the degenerated (robust) value equals brute-force evaluation, and at $(z_0, T) = (\frac{3}{2}, 2)$ every blind sequence loses a branch, so the deficit is at least the minimal mass;
- **S4** the recursion matches the closed form on all 48 cells and horizons $k \leq 4$;
- **S5** the partition coincidence holds cell by cell;
- **S6** the witness-family change sits exactly at $z_0 = 2$ at the one-step horizon and at $z_0 = 1 + k$ at horizon $k \leq T_{\text{obs}}$ (in particular *not* at $z_0 = 2$ for $k = 2$);
- **S7** the deficit is exactly $\frac{1}{2}$ on all 42 nonviable cells;
- **S8** the unrestricted-class table and the sharpened class declaration (coincidence at $k = 1$; difference exactly on the timing cells for $k \geq 2$), by brute force over all sequences;
- **S9** rationality: all 384 campaign alpha-vectors exact rationals (here all in $\{0, 1\}$, maximal denominator 1);
- **S10** deficit monotonicity at every probed belief of both instances;
- **S11** the oscillation identity: branch positions under the alternating sequence, minimum exactly $z_0 - 1$, never exiting for $z_0 \geq 2$ at every horizon to 24;
- **S12** model completeness: transition rows sum to one, $\perp$ is absorbing with observation $y_\perp$, the observation laws sum to one, and witness masking is preserved;
- **S13** the selector split: attaining sets nonempty everywhere probed; value-one level sets nonempty exactly where $V_k(b) = 1$;
- **S14** dimension consistency: the two-floor witnesses pad to $(1, 0, 0), (0, 1, 0)$ on $\mathcal{V} \cup \{\perp\}$ — the first edition suppressed the zero $\perp$ -coordinate in displays, now stated;
- **S15** the jump loci $z_0 = 1 + \min(k, T_{\text{obs}})$ with one-sided values $\frac{1}{2}, 1$ exactly.

On complexity: the worst-case witness growth is exponential ($|\Gamma_{k+1}| \leq |A| |\Gamma_k|^{|Y|}$); the audited instances stay two-witness precisely because the blind window emits a single observation ($|Y| = 1$), so each backup enumerates $|A|$ candidates and deduplicates by exact equality — 384 vectors across the campaign, all $\{0, 1\}$ -valued. No pruning beyond deduplication was needed on the audited classes.

7. Delimitations and outlook

The semantics here is stochastic expectation: the value is a survival *probability*, and the robust quantifiers of the companion papers are recovered only at the deterministic degeneration, which is a change of operator (Remark 1); the stochastic and robust readings of a genuinely stochastic plant differ, and this paper does not conflate them. The timing layer’s statements, here and in the companion papers, are class-scoped: the closed forms are hold-class (sampled-review) values, and the unrestricted sequential class is quantified separately in Theorem 2(e) and Remark 2. The alpha-set growth is exponential in the worst case; the agreement theorem’s classes are two-witness instances precisely because the audited systems are. General information structures beyond history partitions (control-dependent observations, active sensing), continuous-time exit-time value functions, and the five-layer successor architecture are the next items of the programme; the deficit-bound reading of Proposition 3 is the interface through which this theory attaches to the certificate families already established.

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Declarations

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Data availability. No external data were used. **Code availability.** The verification script `paper2_stochastic_selector_v2_verify.py` (exact rational arithmetic; fifteen check families re-deriving Section 6) is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex` in the repository). **AI declaration.** GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review of the first edition; the joint audit of the first edition and this revision were prepared with further AI assistance, under the programme’s verification discipline.